3.33 Affine Algebras under Base Change

This section covers what happens to a k-algebra A under base change, that is $A_{(L)}$ for L/k a field extension. Principally given an integral k-algebra A we would like to ensure that $A_{(L)}$ is reduced (which is related to separability), irreducible and therefore integral. We follow [Stacks] but large parts of exposition are from [Bou89].

Definition 3.33.1 (Geometrically Reduced / Integral Algebra)
A k-algebra A is

- ***geometrically integral*** *(resp.* ***irreducible***, ***reduced****) if the ring $A \otimes_k L$ is* ***integral*** *(resp.* ***irreducible***, ***reduced****) for every field extension L/k.*
- ***algebraically integral*** *(resp.* ***irreducible***, ***reduced****) if the ring $A \otimes_k \overline{k}$ is* ***integral*** *(resp.* ***irreducible***, ***reduced****).*

A few remarks about the important results and notation in the literature

- *As in (3.3.66) we see that* ***integral*** $\iff$ ***irreducible*** *and* ***reduced****.*
- *These are shown to be equivalent (3.33.27), (3.33.37) and (3.33.45). In fact it is sufficient to check the properties for all finite extensions.*
- *[Liu06] uses the "algebraic" form*
- *[Bou89] uses the term "separable" for "geometrically reduced", as this coincides with the notion of separable for algebraic extensions (3.33.20)*
- *Similarly [Bou89] and [Lan11] use the term "regular" for geometrically integral*

We make implicit use of the following results throughout

Proposition 3.33.2
Let A, B be k-algebras, then the structural morphisms $A \to A \otimes_k B$ and $B \to A \otimes_k B$ are injective. Furthermore if A' and B' are k-subalgebras then the canonical homomorphism

$$A' \otimes_k B' \to A \otimes_k B$$

is injective.

Proof. This is (3.4.30) and (3.4.41) ∎

Proposition 3.33.3 (A Subspace is Recovered by Intersection After Base Change)
Let $V \subseteq W$ be k-vector spaces and k'/k a field extension. Then, identifying W with its image in $W \otimes_k k'$ via $w \mapsto w \otimes 1$ (injective, (3.4.31)),

$$(V \otimes_k k') \cap W = V$$

inside $W \otimes_k k'$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

"$\supseteq$" is clear. For "$\subseteq$": choose a k-subspace complement V' of V in W (3.3.139) (k being a field), so $W = V \oplus V'$; tensoring, $W \otimes_k k' \cong (V \otimes_k k') \oplus (V' \otimes_k k')$ (3.4.21). Given $w \in W$ with $w \otimes 1 \in V \otimes_k k'$: writing $w = v + v'$ ($v \in V$, $v' \in V'$) for its decomposition, $w \otimes 1 = (v \otimes 1) + (v' \otimes 1)$ under this direct sum splitting; as $w \otimes 1$ lies in the first summand, the second component vanishes, that is $v' \otimes 1 = 0$; as $v' \mapsto v' \otimes 1$ is injective (3.4.31), $v' = 0$, giving $w = v \in V$. ∎

Proposition 3.33.4
Let A, B be k-algebras. Let $\{a_i\}_{i \in I}$ be a k-linearly independent subset (resp. basis) of A and $\{b_j\}_{j \in I}$ be a k-linearly independent subset (resp. basis) of B. Then $\{a_i \otimes b_j\}_{(i,j) \in I \times J}$ is a k-linearly independent subset (resp. basis) of $A \otimes_k B$.

Furthermore $\{a_i \otimes 1\}_{i \in I}$ is a B-linearly independent subset (resp. basis) of $A \otimes_k B$.

Proof. The first statement is (3.4.31). For the last statement extend $\{a_i\}_{i \in I}$ to a basis (2.3.7) and apply (3.4.26). ∎

Proposition 3.33.5
Let A, B be k-algebras. Then $A \otimes_k B$ is the union of subrings of the form a) $A \otimes_k B'$, b) $A' \otimes_k B$ or c) $A' \otimes_k B'$ where A' and B' are finitely-generated k-subalgebras of A and B respectively.

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If K/k is an algebraic field extension then $A \otimes_k K$ is the union of subrings of the form $A \otimes_k k'$ where k'/k are finite subextensions.

Proof. An arbitrary element $z \in A \otimes_k B$ is a finite sum of elementary tensors. Therefore the result follows from (3.33.2).

The final statement follows from recalling finitely generated algebraic extensions are finite (3.17.44). ∎

Proof. ∎

3.33.1 Etale Algebras

Definition 3.33.6
*Let A be a finite-dimensional k-algebra. Then define the **separable degree** as follows*

$$[A:k]_s := \#\operatorname{AlgHom}_k\left(A, \overline{k}\right)$$

*We say that A is **etale** if this equals the dimension of A as a k-vector space*

$$[A:k]_s = [A:k]$$

*Recall a finite field extension K/k is **etale** if and only if it is separable (3.17.80).*

Lemma 3.33.7
Let A be a finite-dimensional k-algebra and K/k a field extension. Then

$$[\operatorname{Hom}_k(A,K):K] = [A^\vee : k] = [A:k]$$

More precisely if $\{a_1, ..., a_n\}$ is a k-basis for A then $\{a_1^\vee, ..., a_n^\vee\}$ is a K-basis for $\operatorname{Hom}_k(A,K)$.

Proof. This is simply a special case of (3.4.27). ∎

Corollary 3.33.8 (Separable degree is finite)
Let A be a finite-dimensional k-algebra then $\#\operatorname{AlgHom}_k(A,K) \leq [A:k]$ for every field extension K/k.

Proposition 3.33.9 (Image of A is finite-dimensional)
Let A be a finite-dimensional k-algebra and let $N_A \subset \overline{k}$ be the compositum of the images $\varphi(A)$ of all k-algebra homomorphisms $\varphi : A \to \overline{k}$. Then N_A/k is a finite normal extension, and for every field extension Ω/k containing a copy of N_A, that is with a k-embedding $\iota : N_A \to \Omega$, composition with ι is a bijection

$$\operatorname{AlgHom}_k(A, N_A) \to \operatorname{AlgHom}_k(A, \Omega)$$

In particular if Ω contains $\overline{k}$ then these are both equal to $\operatorname{AlgHom}_k\left(A, \overline{k}\right)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\varphi : A \to \overline{k}$ be a k-algebra homomorphism. Then $\varphi(A)$ is a quotient of A and so finite-dimensional over k, and it is a domain, being a subring of a field. A finite-dimensional domain is a field (3.17.47). Therefore $\varphi(A) \subset \overline{k}$ is a finite extension of k. There are only finitely many such φ (3.33.8), say $\varphi_1, ..., \varphi_r$, so $N_A = k(\varphi_1(A), ..., \varphi_r(A))$ is finite over k (3.17.48).

To see N_A/k is normal, let $\sigma \in \operatorname{Mor}_k\left(N_A, \overline{k}\right)$. For each j the composite $\sigma \circ \varphi_j : A \to \overline{k}$ is again a k-algebra homomorphism so $\sigma\left(\varphi_j(A)\right) \subset N_A$, whence $\sigma(N_A) \subset N_A$. As σ is injective and k-linear and N_A is finite-dimensional, $\sigma(N_A) = N_A$. So N_A/k is normal by (3.17.69).

Injectivity of the given map is clear as ι is injective. For surjectivity let $\psi : A \to \Omega$ be a k-algebra homomorphism. As above $L := \psi(A)$ is a finite extension of k, so by (3.17.63) there is a k-embedding $\sigma : L \to \overline{k}$. Then $\sigma \circ \psi : A \to \overline{k}$ has image in N_A by definition. Let $b \in L$ with minimal polynomial $m_{b,k}$. Then $m_{b,k}$ has the root $\sigma(b)$ in N_A and hence splits completely in N_A as N_A/k is normal, say $m_{b,k} = \prod_i (X - c_i)$ with $c_i \in N_A$. Applying ι gives $m_{b,k} = \prod_i (X - \iota(c_i))$ in $\Omega[X]$ and since Ω is a domain the root b is one of the $\iota(c_i)$. Thus $\psi(A) \subseteq \iota(N_A)$ and $\psi = \iota \circ \psi'$ for some $\psi' : A \to N_A$.

Finally $N_A \subset \overline{k}$, so applying this to $\Omega = \overline{k}$ with the inclusion, and then to any $\Omega \supseteq \overline{k}$, shows $\operatorname{AlgHom}_k(A, N_A) = \operatorname{AlgHom}_k\left(A, \overline{k}\right) = \operatorname{AlgHom}_k(A, \Omega)$. ∎

Proposition 3.33.10 (Etale-ness is preserved under base extension)
Let A be a finite-dimensional k-algebra and K/k a field extension then

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$$[A \otimes_k K : K] = [A : k]$$
$$[A \otimes_k K : K]_s = [A : k]_s$$

In particular A is etale if and only if $A \otimes_k K$ is.

Proof. The first equality follows from (3.4.26) and (3.3.109).

For the second equality we see that by (3.33.9) and (3.4.38)

$$\mathrm{AlgHom}_k\big(A, \overline{k}\big) = \mathrm{AlgHom}_k\big(A, \overline{K}\big) \cong \mathrm{AlgHom}_K\big(A \otimes_k K, \overline{K}\big)$$

■

Lemma 3.33.11
Let $A = A_1 \times \cdots \times A_d$ be a finite product of finite-dimensional k-algebras. Then A is etale if and only if each A_i is.

Proof. This follows because

$$\mathrm{AlgHom}_k\big(A_1 \times \cdots \times A_d, \overline{k}\big) \cong \mathrm{AlgHom}_k\big(A_1, \overline{k}\big) \times \cdots \times \mathrm{AlgHom}_k\big(A_d, \overline{k}\big)$$

and a similar consideration for the dimension over k. ■

Lemma 3.33.12
Let A be a finite-dimensional k-algebra. The following are equivalent

a) A is reduced
b) $A \cong L_1 \times \cdots \times L_n$ for some finite extensions L_i/k

Further A is etale if and only if each L_i/k is separable. If k is perfect then A is automatically etale.

Proof. One direction is obvious. Suppose conversely that A is reduced. If $\dim_k A = 1$ then any proper ideal must have strictly lower dimension and therefore A is a field. It is therefore sufficient to show (by induction on dimension) that if A is not a field then $A \cong A_1 \times A_2$ for two (non-zero) k-algebras A_1, A_2.

Suppose A is not a field and let $\mathfrak{a}$ be a non-zero proper ideal of A which has minimal dimension over k as a vector space. As A is reduced, and by minimality, we have $\mathfrak{a}^2 = \mathfrak{a}$. By Nakayama's Lemma (3.19.5) there is $e \in \mathfrak{a}$ such that $ex = x$ for all $x \in \mathfrak{a}$ and in particular $e^2 = e$ and $\mathfrak{a} = Ae$. By assumption $e \neq 0, 1$. Define $f := 1 - e \neq 0, 1$ then evidently $f^2 = f$ and $ef = 0$. The homomorphism of A-modules

$$A \to Ae \times Af$$
$$a \to (ae, af)$$

is injective because $ae + af = a$ and surjective because the image of $a_1 e + a_2 f$ is $(a_1 e, a_2 f)$. The sub-modules Ae and Af are actually sub-rings by the idempotence property and the given map is an isomorphism of rings.

The last statement follows from (3.33.11) and (3.17.106). ■

Lemma 3.33.13 (Primes of a Finite Product of Fields)
Let $A \cong L_1 \times \cdots \times L_n$ be a finite product of fields (in particular, any reduced finite-dimensional k-algebra, (3.33.12)). Then every prime ideal of A is maximal, and there are exactly n of them.

Proof. For each j, $\mathfrak{m}_j := \ker(\pi_j)$ (the kernel of the j-th projection $\pi_j : A \twoheadrightarrow L_j$) is maximal, being the kernel of a surjection onto a field (3.3.59), and the $\mathfrak{m}_j$ are pairwise distinct (the element with 1 in coordinate j and 0 elsewhere lies in every $\mathfrak{m}_i$, $i \neq j$, but not $\mathfrak{m}_j$), giving n maximal ideals.

Let $\mathfrak{p}$ be any prime. As $\bigcap_j \mathfrak{m}_j = \ker(\pi_1, ..., \pi_n) = 0 \subseteq \mathfrak{p}$, (3.3.40) gives $\mathfrak{m}_j \subseteq \mathfrak{p}$ for some j; as $\mathfrak{m}_j$ is already maximal, $\mathfrak{p} = \mathfrak{m}_j$. ■

Lemma 3.33.14 (Algebra Homomorphisms out of a Product of Fields)
Let Ω be a field. The Ω-algebra homomorphisms $\chi : \Omega^n \to \Omega$ are exactly the n projections π_i, and these are pairwise distinct. In particular $\#\,\mathrm{AlgHom}_\Omega(\Omega^n, \Omega) = n$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Indeed let e_i be the i-th standard idempotent, so $e_i^2 = e_i$, $e_i e_j = 0$ for $i \neq j$ and $\sum_i e_i = 1$. Then each $\chi(e_i)$ is an idempotent of the field Ω so equals 0 or 1; as $\chi(e_i)\chi(e_j) = 0$ for $i \neq j$ at most one of them is 1, and as $\sum_i \chi(e_i) = 1$ exactly one, say $\chi(e_j)$, is. Then by Ω-linearity $\chi(x) = \sum_i x_i \chi(e_i) = x_j$, that is $\chi = \pi_j$. The π_i are pairwise distinct as $\pi_i(e_i) = 1 \neq 0 = \pi_j(e_i)$ for $j \neq i$. ■

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Lemma 3.33.15
Let A be a finite-dimensional k-algebra and Ω/k a field extension containing a copy of the extension N_A/k of (3.33.9) (for instance any Ω containing $\overline{k}$). Then the following are equivalent

a) A is etale
b) the map $\Theta : A \otimes_k \Omega \to \Omega^{\mathrm{AlgHom}_k(A,\Omega)}, \quad a \otimes c \mapsto (c\,\varphi(a))_\varphi$ is an isomorphism of Ω-algebras
c) $A \otimes_k \Omega \cong \Omega^n$ as an Ω-algebra for some n

In this case the n of c) equals $[A : k]$, and $A \otimes_k \Omega$ has exactly n maximal ideals, namely the kernels of the coordinates of Θ.

Proof. *a)* $\Longrightarrow$ *b)* By (3.33.9) $\#\,\mathrm{AlgHom}_k(A,\Omega) = \#\,\mathrm{AlgHom}_k\left(A,\overline{k}\right) =: [A : k]_s = [A : k]$ with the last equality because A is assumed etale. Then defining $B := A \otimes_k \Omega$

$$\#\,\mathrm{AlgHom}_\Omega(B,\Omega) \overset{(3.4.38)}{=} \#\,\mathrm{AlgHom}_k(A,\Omega) = [A : k] \overset{(3.4.16)}{=} [B : \Omega] = [B^\vee : \Omega]$$

By (3.17.133) and (3.3.127) we see that $\mathrm{AlgHom}_\Omega(B,\Omega)$ is a Ω-basis for $B^\vee$. By (3.4.38) the map sending $\varphi \in \mathrm{AlgHom}_k(A,\Omega)$ to $\tilde{\varphi} : B \to \Omega$, $a \otimes c \mapsto c\,\varphi(a)$, is a bijection $\mathrm{AlgHom}_k(A,\Omega) \to \mathrm{AlgHom}_\Omega(B,\Omega)$. Write $\mathrm{AlgHom}_k(A,\Omega) = \{\varphi_1, ..., \varphi_n\}$, so that $\tilde{\varphi}_1, ..., \tilde{\varphi}_n$ is the basis of $B^\vee$. Then by (3.3.109) this corresponds to a basis $e_1, ..., e_n$ of B and the algebra isomorphism is exhibited explicitly by

$$\begin{aligned} \Theta : B &\cong \Omega^n \\ a \otimes c &\to (c\,\varphi_1(a), ..., c\,\varphi_n(a)) \end{aligned}$$

Indeed $\Theta(b) = (\tilde{\varphi}_1(b), ..., \tilde{\varphi}_n(b))$ is an Ω-algebra homomorphism, and it is injective since a b killed by every element of a basis of $B^\vee$ is killed by every functional, so $b = 0$. It is therefore an isomorphism by dimensions.

b) $\Longrightarrow$ *c)* Immediate, with $n := \#\,\mathrm{AlgHom}_k(A,\Omega)$, which is finite (3.33.8).

c) $\Longrightarrow$ *a)* By (3.33.14) the Ω-algebra homomorphisms $\Omega^n \to \Omega$ are the n projections π_i, so $\#\,\mathrm{AlgHom}_\Omega(B,\Omega) = n = [B : \Omega]$. As before then $[A : k]_s = \#\,\mathrm{AlgHom}_k\left(A,\overline{k}\right) = \#\,\mathrm{AlgHom}_k(A,\Omega) = \#\,\mathrm{AlgHom}_\Omega(B,\Omega) = [B : \Omega] = [A : k]$ which shows that A is etale.

Finally Ω^n is a finite product of fields, so by (3.33.13) its primes are the n coordinate kernels, and these correspond to the maximal ideals of $A \otimes_k \Omega$ under Θ. ∎

Corollary 3.33.16 (A Field Splits Completely Over Any Normal Extension it Embeds Into)
Let K/k be a finite separable field extension and L/k a field extension containing (a copy of) a normal closure of K/k. Fix a primitive element θ for K/k (3.17.108), with minimal polynomial $m \in k[X]$ of degree $n := [K : k]$, and write $m = \prod_{i=1}^n (X - \beta_i)$ for its (pairwise distinct) roots $\beta_i \in L$. Then there is a canonical isomorphism of L-algebras

$$K \otimes_k L \cong L^n, \quad p(\theta) \otimes c \mapsto (c\,p(\beta_1), ..., c\,p(\beta_n)) \quad (p \in k[X], c \in L)$$

In particular $K \otimes_k L$ has exactly n maximal ideals, $\mathfrak{m}_i$ the kernel of the i-th coordinate.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As K/k is separable it is etale (3.33.6). By (3.17.85).c) applied with $\Omega = \overline{k}$, N_K is a normal closure of K/k, so L contains a copy of N_K (3.17.75) and the isomorphism Θ of (3.33.15) is defined. By (3.17.85).b), $\varphi \mapsto \varphi(\theta)$ is a bijection $\mathrm{Mor}_k(K,L) \to \{\beta_1, ..., \beta_n\}$, and under this indexing Θ sends $p(\theta) \otimes c$ to $(c\,p(\beta_i))_i$. The statement about maximal ideals is the last statement of (3.33.15). ∎

Proposition 3.33.17 (Etale Criteria)
Let A be a finite-dimensional k-algebra. The following are equivalent

a) A is etale
b) A is geometrically reduced
c) $A \otimes_k \overline{k}$ is reduced
d) $A \otimes_k K$ is reduced for some perfect field extension K/k
e) $A \cong L_1 \times \cdots \times L_n$ for L_i/k finite separable extensions

In particular A is reduced.

Proof. *a)* $\Longrightarrow$ *b)* Consider a field extension L/k and the tower of extensions $\overline{L}/L/k$. Then by (3.33.15) $A \otimes_k \overline{L} \cong \overline{L}^n$ is evidently reduced. By (3.33.2) $A \otimes_k L \subset A \otimes_k \overline{L}$ is a subring and therefore also reduced.

b) $\Longrightarrow$ *c)* immediate.

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$c) \implies d)$ immediate as $\overline{k}$ is perfect.

$d) \implies a)$ By (3.33.12) $A \otimes_k K$ is an etale K-algebra so by (3.33.10) A is etale.

$e) \implies a)$ Follows immediately from (3.33.11).

$a) \implies e)$ We know that A is reduced so the result follows from (3.33.12). ∎

Corollary 3.33.18
Let A be a finite-dimensional k-algebra and k perfect. Then the following are equivalent

a) A is etale
b) A is reduced
c) A is geometrically reduced
d) $A \otimes_k \overline{k}$ is reduced.

For completeness we summarize the relationship between the different notions of separability

Corollary 3.33.19 (Equivalent definitions of separability)
Let K/k be a finite extension. Then the following are equivalent

a) K/k is separable algebraic (3.17.53)
b) K/k is etale
c) K/k is geometrically reduced
d) $K \otimes_k \overline{k}$ is reduced

When k is perfect every field extension is separable.

Proof. We have already shown $b) \iff c) \iff d)$. Furthermore $a) \iff b)$ is (3.17.80). ∎

Corollary 3.33.20 (Equivalent definitions of separability)
Let K/k be an algebraic extension. Then the following are equivalent

a) K/k is separable algebraic (3.17.53)
b) Every finite subextension of K/k is etale
c) K/k is geometrically reduced
d) $K \otimes_k \overline{k}$ is reduced

In particular an algebraic extension of a perfect field k is geometrically reduced.

Proof. This follows directly from (3.33.19) because each property has "finite character", i.e. holds if and only if it holds for every finite subextension.

The last statement follows from (3.17.106). ∎

3.33.2 Geometrically Reduced Algebras

Reference is [Bou89] (Section V.15.4). The main results of this section are (3.33.27) and (3.33.29). Observe that if k is perfect every extension K/k satisfies Maclane's Criterion and is therefore Geometrically Reduced.

Lemma 3.33.21
Let A be a reduced k-algebra. Define the canonical map

$$i : A \hookrightarrow \prod_{\mathfrak{p}} k(\mathfrak{p})$$

where $\mathfrak{p}$ runs over all prime ideals of A. Then i is injective. Furthermore for any k-algebra B we have a canonical embedding

$$A \otimes_k B \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k B)$$

Proof. Evidently $\ker(i) = \bigcap \mathfrak{p}$ which by (3.3.46) is (0). The final statement follows from considering the maps

$$A \otimes_k B \overset{i \otimes 1_B}{\hookrightarrow} \left(\prod_{\mathfrak{p}} k(\mathfrak{p}) \right) \otimes_k B \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k B)$$

The first map is injective by (3.4.25) and the second by (...). ∎

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Proposition 3.33.22
Let A be a reduced k-algebra and B a geometrically reduced k-algebra, then $A \otimes_k B$ is reduced.

Proof. By the previous Lemma we have an embedding

$$A \hookrightarrow \prod (k(\mathfrak{p}) \otimes_k B)$$

As B is geometrically reduced this shows that $A \otimes_k B$ is reduced. ∎

Corollary 3.33.23
Let k be a perfect field and A, B reduced k-algebras. Then $A \otimes_k B$ is reduced.

Lemma 3.33.24
Let A be a geometrically reduced k-algebra and K the total ring of fractions. Then K is geometrically reduced.

Conversely if A is integral and K is geometrically reduced, then A is geometrically reduced.

Proof. Let L/k be a field extension and consider a nilpotent element $x \in K \otimes L$. Then by definition

$$x = \sum_{i=1}^{n} (a_i s_i^{-1}) \otimes \lambda_i \quad a_i \in A, s_i \notin Z(A), \lambda_i \in L$$

Let $s = s_1 ... s_n$ then $x \cdot (s \otimes 1) \in A \otimes L \subset K \otimes L$. Then $x \cdot (s \otimes 1)$ is nilpotent and so by definition zero. As $(s \otimes 1)$ is invertible we see $x = 0$ as required.

For the converse $A \subset K$ and therefore $A \otimes_k L \subset K \otimes_k L$. ∎

Lemma 3.33.25
Let K/k be a separably generated field extension. Then K/k is geometrically reduced.

Proof. Consider first the case $K = k(\mathcal{B})$ is purely transcendental. Then for L/k we have $k[\mathcal{B}] \otimes L \cong L[\mathcal{B}]$ is evidently reduced. We may then demonstrate directly that $k(\mathcal{B}) \otimes_k L$ is reduced.

For the general case

$$K \otimes_k L \cong K \otimes_{k(\mathcal{B})} (k(\mathcal{B}) \otimes_k L)$$

We have shown that $k(\mathcal{B}) \otimes_k L$ is reduced and by (3.33.20) $K/k(\mathcal{B})$ is geometrically reduced as by assumption it is separable. By (3.33.22) $K \otimes_k L$ is reduced and K is geometrically reduced as required. ∎

Lemma 3.33.26
Let A be a reduced k-algebra for which k is perfect. Then A is geometrically reduced.

Proof. We prove the result in increasing generality

a) We first consider the case $A = K/k$ is a finitely generated field extension. By (3.17.151) K/k is separably generated and so by (3.33.25) it is geometrically reduced.
b) When A is a finitely-generated k-algebra by (3.33.21) there is a canonical embedding

$$A \otimes_k L \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k L)$$

By assumption $k(\mathfrak{p})$ is finitely generated and so by the first part is geometrically reduced. Therefore $A \otimes_k L$ is reduced as required.
c) The general case follows from b) and the reduction to the finitely generated case (3.33.5).

∎

Proposition 3.33.27 (Maclane's Criterion for Algebras)
Let A be a k-algebra. Then the following are equivalent

a) A is geometrically reduced
b) $A \otimes_k \overline{k}$ is reduced
c) $A \otimes_k k^{p^{-\infty}}$ is reduced
d) $A \otimes_k K$ is reduced for some perfect extension K/k
e) $A \otimes_k k^{p^{-1}}$ is reduced
f) $A \otimes_k k'$ is reduced for every finite subextension k'/k of $k^{p^{-1}}/k$
g) The following k-module homomorphism

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$$\begin{aligned} m : A \otimes_k k^{p^{-1}} &\to A \\ a \otimes \lambda &\to \lambda^p a^p \end{aligned}$$

is injective (that is $k^{p^{-1}}$ and A are linearly disjoint with respect to the p-th power embedding into A)

h) A satisfies Maclane's Criterion (3.17.147)

In particular when k is perfect then this is equivalent to A being reduced.

Proof. The case $p = 1$ is simply (3.33.26) because $k^p = k = k^{p^{-1}} = k^{p^{-\infty}}$ is perfect and g) is equivalent to A being reduced. Therefore we assume $p > 1$.

a) $\implies$ *b)* By definition.

b) $\implies$ *c)* By (3.17.63) there is an embedding $k^{p^{-\infty}} \hookrightarrow \overline{k}$ whence an embedding $A \otimes_k k^{p^{-1}} \hookrightarrow A \otimes_k \overline{k}$.

c) $\implies$ *d)* By (3.17.103) $k^{p^{-\infty}}$ is perfect.

d) $\implies$ *e)* By (3.17.104) there is an embedding $k^{p^{-1}} \hookrightarrow K$ whence an embedding $A \otimes_k k^{p^{-1}} \hookrightarrow A \otimes_k K$.

e) $\implies$ *f)* This is immediate as we have an embedding $A \otimes_k k' \hookrightarrow A \otimes_k k^{p^{-1}}$.

f) $\iff$ *g)* Suppose $x := \sum_i \lambda_i \otimes a_i \in A \otimes_k k^{p^{-1}}$ is a finite sum then we may consider the finite subextension $k' := k\left(\{a_i\}_i\right)$. Observe that

$$x^p = \sum_i \lambda_i^p \otimes a_i^p = \sum_i 1 \otimes \lambda_i^p a_i^p = 1 \otimes \left(\sum_i \lambda_i^p a_i^p \right) =: 1 \otimes m(x)$$

Therefore we see $A \otimes_k k'$ is reduced for every such k' iff $m(x)$ is injective.

g) $\iff$ *h)* Consider the Frobenius morphisms $i : k^{p^{-1}} \to A$ and $j : A \to A$. Then g) is equivalent to $k^{p^{-1}}$ and A being linearly disjoint in A with respect to i, j and the result follows from the characterization of linear disjointness (3.32.55).

h) $\implies$ *a)* Let $\{a_i\}_{i \in I}$ be a k-basis for A then by (3.4.26) $\{a_i \otimes 1\}_{i \in I}$ is an L-basis for $A \otimes_k L$. Given $x \in A \otimes_k L$ we have

$$x = \sum_{i \in I} a_i \otimes \lambda_i \quad \lambda_i \in L$$

and

$$x^p = \sum_{i \in I} \lambda_i^p \otimes a_i^p = \sum_{i \in I} \lambda_i^p (a_i^p \otimes 1)$$

By assumption $\{a_i^p\}_{i \in I}$ is k-linearly independent and therefore $\{a_i^p \otimes 1\}_{i \in I}$ is L-linearly independent. Therefore $x^p = 0 \implies \lambda_i^p = 0$ for all $i \in I \implies \lambda_i = 0$ for all $i \in I \implies x = 0$, as L is reduced. It follows from (3.17.148) that $A \otimes_k L$ is reduced, as we assumed that $p > 1$. ∎

Corollary 3.33.28 (Maclane's Criterion for Fields)
Let K/k be a field extension with characteristic exponent p. Then the following are equivalent

a) K/k is geometrically reduced
b) $K \otimes_k \overline{k}$ is reduced
c) $K \otimes_k k^{p^{-1}}$ is reduced
d) $K \otimes_k k^{p^{-\infty}}$ is reduced
e) $K \otimes_k k'$ is reduced for some perfect extension k'/k
f) K and $k^{p^{-1}}$ are linearly disjoint (with respect to some common field extension)
g) $K \otimes_k k^{p^{-1}}$ is an integral domain
h) $K \otimes_k k^{p^{-1}}$ is a field
i) K/k satisfies Maclane's Criterion (3.17.147)

Observe i) has finite character *in the sense it holds if and only if it holds for every finitely-generated subextension.*

In particular when k is perfect every extension is geometrically reduced.

Proof. By (3.33.27) a)-f), i) are equivalent (using (3.32.56) to generalise the linear disjoint condition to any ambient field) and by (3.32.56) again f)-h) are equivalent. ∎

Proposition 3.33.29 (Separable Field Extensions)
Let K/k be a field extension. Then the following are equivalent

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a) K/k is geometrically reduced
b) $K \otimes_k \overline{k}$ is reduced
c) K/k satisfies Maclane's Criterion (3.17.147)
d) Every finitely generated subextension K/k is separably generated

In particular if k is perfect every field extension satisfies these properties.

Proof. By (3.33.28) we have $a) \iff b) \iff c)$. Further $c) \implies d)$ is Lemma (3.17.151), since every subextension satisfies Maclane's Criterion. For $d) \implies a)$ observe that every finitely generated subextension is geometrically reduced by (3.33.25). We may use (3.33.2) to argue that K/k is geometrically reduced. ∎

Corollary 3.33.30
Let K/k be a finitely generated field extension. Then the following are equivalent.

a) K/k is geometrically reduced
b) $K \otimes_k \overline{k}$ is reduced
c) K/k satisfies Maclane's Criterion (3.17.147)
d) Every finitely generated subextension of K/k is separably generated
e) K/k is separably generated

Proof. We've already shown equivalence of a)-d) and clearly $d) \implies e)$. Finally $e) \implies a)$ is simply (3.33.25). ∎

3.33.3 Geometrically Irreducible Algebras

The reference for this section is [Stacks] (Tag 00I2) presented here with a more elementary proof of (3.33.35).

Lemma 3.33.31
Let A, B be k-algebras. If $a \otimes 1 \in A \otimes_k B$ is a zero-divisor then so is a.

Proof. Consider a basis $\{a_i\}_{i\in I}$ for A such that $a_{i_0} = a$. If a is not a zero-divisor then we may show directly that $\{aa_i\}_{i\in I}$ is linearly independent. Then by extending to a basis of A we may use (3.4.26) to show that $\{(aa_i) \otimes 1\}_{i\in I}$ is B-linearly independent. We see immediately that $a \otimes 1$ is not a zero-divisor as required. ∎

Lemma 3.33.32
Let A be a ring and $\mathfrak{a} \subset N(A)$ a locally nilpotent ideal. Then A is irreducible iff $A/\mathfrak{a}$ is irreducible.

Lemma 3.33.33
Let $A \subset B$ be a subring of an irreducible ring. Then A is irreducible.

In particular if B is a geometrically irreducible k-algebra, then so is any subalgebra.

Proof. By (3.21.7) every minimal prime in A is the contraction of a minimal prime. As there is precisely one minimal prime of B then we are done.

The final statement follows because the canonical map $A \otimes_k L \to B \otimes_k L$ is injective. ∎

When k is algebraically closed then nothing interesting happens.

Lemma 3.33.34
Let A, B be irreducible (resp. integral) k-algebras with k algebraically closed. Then $A \otimes_k B$ is irreducible (resp. integral).

In particular A is a geometrically irreducible (resp. integral) k-algebra.

Proof. Observe that $N(A) \otimes_k B + A \otimes_k N(B) \subset N(A \otimes_k B)$. For $(n \otimes \lambda)^m = n^m \otimes \lambda^m = 0$ and we may use (...) to argue any linear combination of such elements is nilpotent. By (...) we have an isomorphism

$$A/N(A) \otimes_k B/N(B) \cong (A \otimes_k B)/(N(A) \otimes_k B + A \otimes_k N(B))$$

Then by (3.33.32) $A/N(A) \otimes_k B/N(B)$ is irreducible iff $A \otimes_k B$ is. Therefore we may consider only the case A, B reduced and therefore integral, and show that $A \otimes_k B$ is integral and a-fortiori irreducible.

If $A \otimes_k B$ were not integral it would contain a non-trivial zero divisor, and this would also be true for $A' \otimes_k B'$ for some finitely generated subalgebras A' and B'. Therefore we may assume wlog that A and B are finitely generated.

To simplify the argument choose a k-basis $\{b_i\}_{i\in I}$ for B. By (...) $\{1 \otimes b_i\}_{i\in I}$ is an A-basis for $A \otimes_k B$. Therefore if we have a product of elements equal to 0 then may write it as follows

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$$\left(\sum_i a_i \otimes b_i\right)\left(\sum_i a_i' \otimes b_i\right) = 0$$

for some $a_i, a_i' \in A$ all but finitely many zero. Let $\mathfrak{m} \lhd A$ be a maximal ideal, then by the Weak Nullstellensatz (3.32.23) the structural morphism $k \to A/\mathfrak{m}$ is an isomorphism. Reduce the equation to $A/\mathfrak{m} \otimes_k B \cong B$ to find

$$\left(\sum_i \overline{a}_i b_i\right)\left(\sum_i \overline{a}_i' b_i\right) = 0$$

As B is an integral domain, and b_i are linearly independent we find $\overline{a}_i, \overline{a}_i' = 0$. As $\mathfrak{m}$ was arbitrary we find $a_i, a_i' \in \bigcap \mathfrak{m}$ for all i. By assumption (0) is prime and so by the Strong Nullstellensatz (3.34.3) we find $a_i = a_i' = 0$. This shows that $A \otimes_k B$ is integral as required.

The last statement follows because a field K/k is always irreducible. ∎

Proposition 3.33.35 (Extension of Scalars is Flat)
Let A be a k-algebra and $L/K/k$ a tower of field extensions. Then the canonical ring homomorphism

$$A \otimes_k K \to A \otimes_k L$$

satisfies Going Down. Further we have a well-defined surjective map of minimal primes

$$\mathrm{MinPrime}(A \otimes_k L) \to \mathrm{MinPrime}(A \otimes_k K)$$

given by the inverse image.

Proof. By (...) there is a commutative diagram

$$\begin{array}{ccc} A \otimes_k K & \longrightarrow & A \otimes_k L \\ & \searrow & \downarrow \sim \\ & & (A \otimes_k K) \otimes_K L \end{array}$$

Therefore it is enough to show that $A \to A \otimes_k K$ satisfies going down. Suppose $\mathfrak{p}_1 \subseteq \mathfrak{p}_2 = \mathfrak{q}_2^c$. There is a commutative diagram

$$\begin{array}{ccc} A/\mathfrak{p}_1 & \xrightarrow{\alpha} & A/\mathfrak{p}_1 \otimes_k K \\ \uparrow \pi & & \uparrow \pi \otimes 1 \\ A & \xrightarrow{\beta} & A \otimes_k K \end{array}$$

where (3.4.42) shows that $\ker(\pi \otimes 1) = \mathfrak{p}_1^e$.

Let $\tilde{\mathfrak{q}}_1$ be a minimal prime of $A/\mathfrak{p}_1 \otimes_k K$ contained in $\tilde{\mathfrak{q}}_2 := (\pi \otimes 1)(\mathfrak{q}_2)$ by (3.3.44). Define $\mathfrak{q}_1 := (\pi \otimes 1)^{-1}(\tilde{\mathfrak{q}}_1)$ then $\mathfrak{q}_1$ is a prime minimal over $\mathfrak{p}_1^e$. As $\pi \otimes 1$ is surjective it is contained in $\mathfrak{q}_2$. We require to prove that $\beta^{-1}(\mathfrak{q}_1) = \mathfrak{p}_1$, which we readily see is implied by $\alpha^{-1}(\tilde{\mathfrak{q}}_1) = (0)$.

By (3.5.46) $\tilde{\mathfrak{q}}_1$ consists of zero divisors and by (3.33.31) so does $\alpha^{-1}(\tilde{\mathfrak{q}}_1)$. As $A/\mathfrak{p}_1$ is an integral domain we see that this equals (0) and we are done.

This immediately shows that the inverse image of a minimal prime is again a minimal prime. By (3.21.7) every minimal prime is a contracted minimal prime. ∎

Lemma 3.33.36 (Purely Inseparable Extensions of Scalars are Homeomorphisms)
Let k be a field with characteristic exponent p, A a k-algebra and k'/k a purely inseparable field extension. Consider the structural algebra homomorphism

$$\varphi : A \to A \otimes_k k'$$

Then φ induces an order-preserving bijection between prime ideals

$$\mathfrak{q}^c \leftarrow \mathfrak{q}$$

In particular A is irreducible iff $A \otimes_k k'$ is.

Furthermore a purely inseparable extension is geometrically irreducible.

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Proof. If $p = 1$ then k'/k is trivial and the statement is obvious. So we may assume wlog that $p > 1$.

If φ were surjective then we would have equality $\mathfrak{q}^{ce} = \mathfrak{q}$ and which would imply the map of prime ideals is injective. However we note that φ is "almost" surjective, namely for every $z = \sum_{i=1}^{N} a_i \otimes \lambda_i$ there exists $n \geq 0$ such that $z^{p^n} \in \varphi(A)$. For k'/k is purely inseparable so by definition there is some $n \geq 0$ such that $\lambda_i^{p^n} \in k$ for all $i = 1...N$. In this case $z^{p^n} = \sum_{i=1}^{N} a_i^{p^n} \lambda_i^{p^n} \otimes 1$ which is in the image of A as required.

This allows us to argue the map is injective for suppose $\mathfrak{p} := \mathfrak{q}^c = (\mathfrak{q}')^c$ and $z \in \mathfrak{q} \setminus \mathfrak{q}'$. Then by the previous argument $z^{p^n} = \varphi(x) \implies z^{p^n} \in \mathfrak{q} \implies z^{p^n} \in \mathfrak{q}' \implies z \in \mathfrak{q}'$ which is a contradiction. Therefore contraction of prime ideals is injective.

Finally we may observe that φ is integral (3.22.22) and injective (3.33.2). Therefore it satisfies the lying over property (3.22.18) and so every prime ideal in A is contracted and the given map is surjective. ∎

The following result shows it is sufficient to ensure that A remains irreducible under a finite separable base change.

Proposition 3.33.37
Let A be a k-algebra. Then the following are equivalent

a) A is geometrically irreducible
b) $A \otimes_k k'$ is irreducible for every finite separable extension k'/k
c) $A \otimes_k k^{\text{sep}}$ is irreducible
d) $A \otimes_k \overline{k}$ is irreducible

Proof. *a)* $\implies$ *b)* immediate

b) $\implies$ *c)* Suppose $\mathfrak{q}_1, \mathfrak{q}_2$ are minimal prime ideals of $A \otimes_k k^{\text{sep}}$. Let $k^{\text{sep}}/k'/k$ be a fixed finite separable extension and consider the subring

$$A \otimes_k k' \subset A \otimes_k k^{\text{sep}}$$

together with the prime ideals $\mathfrak{p}_i := \mathfrak{q}_i \cap (A \otimes_k k')$. By (3.33.35) the prime ideals $\mathfrak{p}_i$ are minimal and therefore $\mathfrak{p}_1 = \mathfrak{p}_2$ by assumption. As k'/k was an arbitrary finite subextension we may conclude by (3.33.5) that $\mathfrak{q}_1 = \mathfrak{q}_2$ and $A \otimes_k k^{\text{sep}}$ is irreducible.

c) $\implies$ *d)* By (...) $A \otimes_k \overline{k} \cong (A \otimes_k k^{\text{sep}}) \otimes_{k^{\text{sep}}} \overline{k}$ and so is irreducible by (3.33.36), since $\overline{k}/k^{\text{sep}}$ is purely inseparable (3.17.101).

d) $\implies$ *a)* Let K/k be a field extension then we may identify $\overline{k}$ and K as subextensions of $\overline{K}/k$. Then $A \otimes_k \overline{K} \cong \left(A \otimes_k \overline{k}\right) \otimes_{\overline{k}} \overline{K}$. By (3.33.34) $A \otimes_k \overline{K}$ is irreducible. Finally from (3.33.35) we see that $A \otimes_k K$ is irreducible as required. ∎

Proposition 3.33.38
Let A be a geometrically irreducible K-algebra and K/k a geometrically irreducible field extension. Then A/k is geometrically irreducible.

Proof. It is sufficient by (3.33.37) to consider base change by a finite separable extension k'/k. Observe by transitivity

$$A \otimes_k k' \cong A \otimes_K (K \otimes_k k')$$

so it would be sufficient to show that $K \otimes_k k'$ is a field. By hypothesis $K \otimes_k k'$ is irreducible. Furthermore k'/k is geometrically reduced (3.33.19) whence $K \otimes_k k'$ is reduced and therefore an integral domain. Using the fact k'/k is algebraic we may conclude from (3.32.56) that $K \otimes_k k'$ is a field, and we are done. ∎

Proposition 3.33.39 (Primary Extension)
Let K/k be a field extension. Then K/k is geometrically irreducible if and only if it is relatively separably closed.

In this case the subfield of algebraic elements is purely inseparable over k. In particular an algebraic extension is geometrically irreducible if and only if it is purely inseparable.

We say such an extension is ***primary****.*

Proof. Suppose that K/k is relatively separably closed. By (3.33.37) it is enough to show that $K \otimes_k k'$ is irreducible for every finite separable extension k'/k.

Let $K/K'/k$ be the relative algebraic closure then K/K' is relatively algebraically closed (3.17.46) and K'/k is purely inseparable because $K_s = k$ by hypothesis (3.17.99). Observe K'/k is geometrically irreducible (3.33.36) so by (3.33.38) it is enough to show that K/K' is geometrically irreducible. This reduces to the case that K/k is relatively algebraically closed.

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As before it is sufficient to consider base change by a finite separable extension k'/k. By (3.17.79) $k' = k(\alpha)$ is simple. Let $m(X)$ be the minimal polynomial for α then $k(\alpha) \cong k[X]/(m)$. From (3.33.40) $m(X)$ remains irreducible in $K[X]$ and therefore (using $k(\alpha) \otimes_k K \cong (k[X] \otimes_k K)/(m)$ as K-algebras, (3.4.42), and $k[X] \otimes_k K \cong K[X]$, (3.7.8))

$$K \otimes_k k(\alpha) \cong K[X]/(m)$$

is a field and in particular irreducible.

Conversely suppose that K/k is geometrically irreducible and $\alpha \in K$ is separable. Then $k(\alpha)$ is geometrically irreducible by (3.33.33). Therefore, using $k(\alpha) \otimes_k \overline{k} \cong \big(k[X] \otimes_k \overline{k}\big)/(m_\alpha)$ as $\overline{k}$-algebras (3.4.42), then $k[X] \otimes_k \overline{k} \cong \overline{k}[X]$ (3.7.8), then by the Chinese Remainder Theorem (3.3.62)

$$k(\alpha) \otimes_k \overline{k} \cong k[X]/(m_\alpha) \otimes_k \overline{k} \cong \overline{k}[X]/(m_\alpha) \cong \overline{k}[X]/(X-\alpha_1)^{n_1} \times \cdots \times \overline{k}[X]/(X-\alpha_r)^{n_r}$$

is irreducible where $\alpha_1, ..., \alpha_r$ are the distinct roots in $\overline{k}$. This implies $r = 1$ and therefore α is purely inseparable (3.17.91). Being both separable and purely inseparable $\alpha \in k$ (3.17.92) as required. ∎

We made use of

Lemma 3.33.40
Let K/k be a field extension such that k is algebraically closed in K. Then every irreducible polynomial $f \in k[X]$ remains irreducible in $K[X]$.

Proof. Consider the extension $\overline{K}$ which contains $\overline{k}$. And suppose $g \mid f$ in $K[X]$. As f splits completely in $\overline{k}$ then by unique factorisation in $\overline{K}[X]$ the coefficients of g lie in $K \cap \overline{k} = k$. Therefore by irreducibility $g = f$ or is constant. In particular f is irreducible in $K[X]$. ∎

Proposition 3.33.41
Let A be an integral k-algebra and $K := \mathrm{Frac}(A)$. Then A is geometrically irreducible iff K is geometrically irreducible.

Proof. Suppose K is geometrically irreducible and consider an arbitrary field extension L/k. Then we have an embedding $A \otimes_k L \hookrightarrow K \otimes_k L$, and therefore an embedding $(A \otimes_k L)_{\mathrm{red}} \hookrightarrow (K \otimes_k L)_{\mathrm{red}}$. By assumption these are integral and we conclude that $(A \otimes_k L)$ is irreducible.

Conversely suppose $A \otimes_k L$ is irreducible then define $\Omega := \mathrm{Frac}\big((A \otimes_k L)_{\mathrm{red}}\big)$. As $L \hookrightarrow (A \otimes_k L)$ is injective and L is reduced we have a canonical map $L \hookrightarrow \Omega$. Similarly we have a canonical map $A \hookrightarrow \Omega$ and therefore a map $K \hookrightarrow \Omega$. Therefore we have a well-defined ring homomorphism

$$\begin{aligned} \psi : K \otimes_k L &\to \Omega \\ ab^{-1} \otimes \lambda &\to (a \otimes \lambda) \cdot (b \otimes 1)^{-1} \end{aligned}$$

We must show that if $x \in \ker(\psi)$ then x is nilpotent. By definition

$$x = \sum_{i=1}^{n} \lambda_i \otimes \mu_i$$

We may find $a \in A$ such that $y = ax$ is of the form

$$y = \sum_{i=1}^{n} a_i \otimes \mu_i \in A \otimes_k L$$

We observe $y \in \ker(A \otimes_k L \to \Omega)$ whence y, and therefore x, is nilpotent. This means we have an embedding

$$(K \otimes_k L)_{\mathrm{red}} \hookrightarrow \Omega$$

which means $(K \otimes_k L)_{\mathrm{red}}$ is integral and therefore $K \otimes_k L$ is irreducible. ∎

Proposition 3.33.42
Let A be an integral k-algebra and $K := \mathrm{Frac}(A)$. Then the following are equivalent

a) A is geometrically irreducible
b) $A \otimes_k k'$ is irreducible for every finite separable extension k'/k
c) $A \otimes_k \overline{k}$ is irreducible
d) K is geometrically irreducible
e) K is relatively separably closed

Proof. We may combine (3.33.37), (3.33.41) and (3.33.39). ∎

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3.33.4 Geometrically Integral Algebras

Clearly a geometrically (resp. algebraically) integral algebra is integral, we are interested in providing sufficient conditions for the reverse implication.

Proposition 3.33.43
Let A be a geometrically integral k-algebra and B an integral k-algebra. Then $A \otimes_k B$ is an integral domain.

Proof. Let $L = \mathrm{Frac}(B)$ then $A \otimes_k B$ is a subring of $A \otimes_k L$ and we are done. ∎

Proposition 3.33.44 (Criteria for Geometrically Integral in terms of Quotient Field)
Let A be an integral k-algebra and $K \coloneqq \mathrm{Frac}(A)$. Then $A \otimes_k L$ is integral if and only if $K \otimes_k L$ is integral.

In particular A is geometrically (resp. algebraically) integral if and only if K is geometrically (resp. algebraically) integral.

Proof. This follows exactly the same lines as (3.33.41). ∎

Proposition 3.33.45 (Criteria to be Geometrically Integral)
Let A be a k-algebra. Then the following are equivalent

a) A is geometrically integral
b) A is geometrically reduced and irreducible
c) $A \otimes_k \overline{k}$ is an integral domain
d) $A \otimes_k k'$ is an integral domain for every finite extension k'/k

Note we may restrict d) to only finite separable, and finite purely inseparable extensions of height at most 1.

Proof. This is largely a collection of existing results. $a) \iff b)$ is by definition and (3.3.67). Then $b) \iff c) \iff d)$ is from the reduced (3.33.27) and irreducible (3.33.37) cases, together with the trivial observation that reduced + irreducible $\iff$ integral (3.3.67). ∎

Proposition 3.33.46 (Regular Extension)
Let K/k be a finitely generated field extension. Then the following are equivalent

a) K/k is geometrically integral
b) K/k is separably generated and relatively separably closed
c) K/k is separably generated and relatively algebraically closed
d) $K \otimes_k k'$ is an integral domain for every finite *extension k'/k*
e) $K \otimes_k k'$ is a field for every algebraic *extension k'/k*
f) $K \otimes_k \overline{k}$ is an integral domain
g) $K \otimes_k \overline{k}$ is a field

*We say such an extension is **regular**.*

Proof. $a) \implies b)$ is (3.33.30) and (3.33.39). $c) \implies b)$ is immediate. For the converse suppose $\alpha \in K/k$ is algebraic then $k(\alpha)/k$ is geometrically reduced (3.33.30) and therefore separable algebraic (3.33.19). By hypothesis $\alpha \in k$ as required, giving $b) \iff c)$. For $b) \implies a)$: K/k being separably generated, it is geometrically reduced (3.33.25), and being relatively separably closed, it is geometrically irreducible (3.33.39) together these give $a)$ (3.3.67). This closes the loop $a) \iff b) \iff c)$. $a) \iff d) \iff f)$ was already proven in (3.33.45), and $f) \iff g)$ was proven in (3.32.56). $e) \implies d), g)$ is immediate.

For $f) \implies e)$ fix an algebraic k'/k and an embedding $k' \hookrightarrow \overline{k}$ (exists by (3.17.63)); the canonical map $K \otimes_k k' \to K \otimes_k \overline{k}$ is injective (3.33.2) (k' being a k-subalgebra of $\overline{k}$), so $K \otimes_k k'$ is a subring of the domain $K \otimes_k \overline{k}$, hence itself a domain; as k'/k is algebraic, this domain is moreover a field (3.32.56). ∎

Corollary 3.33.47
Let A be a finitely-generated integral k-algebra and K the quotient field. Then the following are equivalent

a) A is geometrically integral
b) K/k is geometrically integral
c) K/k is separably generated and relatively separably closed
d) $K \otimes_k k'$ is an integral domain for every finite extension k'/k
e) $K \otimes_k \overline{k}$ is an integral domain
f) $K \otimes_k \overline{k}$ is a field

Proof. $a) \iff b)$ is (3.33.44) and the rest is (3.33.46). ∎

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The mechanism underlying (6.3.56) below (base-changing a place of an algebraic function field along an algebraic extension of the constant field) is really a fact about discrete valuation rings, unrelated to the transcendence degree 1 setting of function fields; we isolate it here.

Proposition 3.33.48 (Unramified Base Extension of a Discrete Valuation Ring)
Let k be a perfect field and $\mathcal{O}$ a discrete valuation ring containing k, with fraction field F and maximal ideal $\mathfrak{m}$, such that F/k is geometrically integral and $k(\mathfrak{m}) := \mathcal{O}/\mathfrak{m}$ is finite over k.

Let k'/k be algebraic, write $\mathcal{O}_{k'} := \mathcal{O} \otimes_k k'$ and $\mathfrak{b} := \mathfrak{m}\mathcal{O}_{k'}$. Then

a) $\mathcal{O}_{k'}$ has exactly r maximal ideals $\mathfrak{p}_1, ..., \mathfrak{p}_r$, the preimages of those of $\mathcal{O}_{k'}/\mathfrak{b}$
b) the localisation $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ is a place of $F_{k'}$ unramified over $\mathcal{O}$: a uniformiser of $\mathcal{O}$ generates the maximal ideal $\mathfrak{p}_j(\mathcal{O}_{k'})_{\mathfrak{p}_j}$, and
c) every valuation ring R of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$, and $R \cap F = \mathcal{O}$ is of the form $R = (\mathcal{O}_{k'})_{\mathfrak{p}_j}$ for some j, and
d) when k'/k is finite then $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ is a discrete valuation ring, and
e) $(\mathcal{O}_{k'})_{\mathfrak{p}_j} \cap F = \mathcal{O}$ and $\mathfrak{p}_j(\mathcal{O}_{k'})_{\mathfrak{p}_j} \cap F = \mathfrak{m}$, inside $F_{k'}$: that is $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ dominates $\mathcal{O}$ and restricts to it. Together with c), the valuation rings R of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$ and $R \cap F = \mathcal{O}$ are therefore exactly the $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let π be a uniformiser of $\mathcal{O}$, so $\mathfrak{m} = (\pi)$ (3.29.2). Since F/k is geometrically integral and k'/k algebraic, $F_{k'}$ is a field (3.33.46), and $\mathcal{O}_{k'}$ is a subring (...) and in particular an integral domain. Further $\mathcal{O}_{k'}$ integral over $\mathcal{O}$ (3.22.22).

$\mathcal{O}_{k'}/\mathfrak{b} = \mathcal{O}_{k'}/\mathfrak{m}\mathcal{O}_{k'} \cong (\mathcal{O}/\mathfrak{m}) \otimes_k k' = k(\mathfrak{m}) \otimes_k k'$ as k-algebras (3.4.42): a finite-dimensional (as $k(\mathfrak{m})/k$ is finite, by hypothesis) reduced ($k(\mathfrak{m})/k$ and k'/k both separable, k being perfect, (3.17.106), (3.33.29)) k-algebra, hence a finite product of r fields (3.33.12), with every prime maximal (3.33.13).

a) Finitely many maximal ideals. $\mathcal{O}_{k'}/\mathfrak{b}$, being a finite product of r fields (above), has finitely many maximal ideals $\mathfrak{p}'_1, ..., \mathfrak{p}'_r$, all its primes maximal (3.33.13). Every maximal ideal of $\mathcal{O}_{k'}$ contracts to a maximal ideal of $\mathcal{O}$ (3.22.16), that is to $\mathfrak{m}$ ($\mathcal{O}$ being local), hence contains $\mathfrak{b}$; so, by the ideal correspondence for $\mathcal{O}_{k'} \twoheadrightarrow \mathcal{O}_{k'}/\mathfrak{b}$ (3.3.56), the maximal ideals of $\mathcal{O}_{k'}$ are exactly the preimages $\mathfrak{p}_1, ..., \mathfrak{p}_r$ of $\mathfrak{p}'_1, ..., \mathfrak{p}'_r$ – and, every prime of the quotient being maximal, these are also *every* prime of $\mathcal{O}_{k'}$ containing $\mathfrak{b}$.

As $\mathfrak{b}$ is radical (the quotient being reduced), the Prime Nullstellensatz (3.3.46) gives $\mathfrak{b} = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r$; as the $\mathfrak{p}_j$ are pairwise comaximal (distinct maximal ideals), this equals $\mathfrak{p}_1 \cdots \mathfrak{p}_r$ (3.3.62).

b) Unramifiedness. Write $\mathcal{O}_j := (\mathcal{O}_{k'})_{\mathfrak{p}_j}$. By (3.5.18).a) and multiplicativity of localisation on ideals (3.28.9),

$$\pi\mathcal{O}_j = \mathfrak{b}\mathcal{O}_j = \mathfrak{b}_{\mathfrak{p}_j} = (\mathfrak{p}_1)_{\mathfrak{p}_j} \cdots (\mathfrak{p}_r)_{\mathfrak{p}_j} = \mathfrak{p}_j\mathcal{O}_j$$

(the first equality as $\mathfrak{b} = \pi\mathcal{O}_{k'}$; the factors $i \neq j$ becoming the unit ideal, as $\mathfrak{p}_i \not\subset \mathfrak{p}_j$). So π generates $\mathfrak{p}_j\mathcal{O}_j$, once $\mathcal{O}_j$ is known to be a valuation ring of $F_{k'}$ properly between k' and $F_{k'}$ – established next.

d) The finite case: $\mathcal{O}_j$ is a DVR. Suppose k'/k is finite. Then $\mathcal{O}_{k'}$ is a finite, free $\mathcal{O}$-module (3.4.26), hence a Noetherian $\mathcal{O}$-module (3.13.4) ($\mathcal{O}$ Noetherian, being a PID (3.14.2)), so Noetherian as a ring (3.13.3) (every ideal being in particular an $\mathcal{O}$-submodule). So $\mathcal{O}_j$ is Noetherian (3.13.7), local with maximal ideal $\mathfrak{p}_j\mathcal{O}_j$ (3.5.42), and a domain (above). It has Krull dimension 1: as $\mathcal{O} \hookrightarrow \mathcal{O}_{k'}$ is injective and integral (module-finite), $\dim \mathcal{O}_{k'} = \dim \mathcal{O} = 1$ (3.22.24) ($\mathcal{O}$ a DVR, hence of dimension 1 (3.29.2)); as $\mathcal{O}_{k'}$ is a domain, $(0) \subsetneq \mathfrak{p}_j$ gives $\mathrm{ht}(\mathfrak{p}_j) \geq 1$, while $\mathrm{ht}(\mathfrak{p}_j) \leq \dim \mathcal{O}_{k'} = 1$, so $\dim \mathcal{O}_j = \mathrm{ht}(\mathfrak{p}_j) = 1$ (3.20.4).b). With $\mathfrak{p}_j\mathcal{O}_j = (\pi)$ (above), $\mathcal{O}_j$ is a DVR (3.29.2).

In general, $\mathcal{O}_j$ is a valuation ring. Write $k' = \bigcup_i k'_i$ as the directed union of its finite subextensions (3.17.49), so $\mathcal{O}_{k'} = \bigcup_i \mathcal{O}_{k'_i}$ for $\mathcal{O}_{k'_i} := \mathcal{O} \otimes_k k'_i$ (a directed union of subrings, via (3.33.2)). For each i, $\mathfrak{p}_{i,j} := \mathfrak{p}_j \cap \mathcal{O}_{k'_i}$ contains $\mathfrak{m}\mathcal{O}_{k'_i}$ (as $\mathfrak{p}_j \supseteq \mathfrak{b}$), so corresponds, under the ideal correspondence for the finite-dimensional ring $\mathcal{O}_{k'_i}/\mathfrak{m}\mathcal{O}_{k'_i} = k(\mathfrak{m}) \otimes_k k'_i$ (as k-algebras, (3.4.42)) (reduced, as $k(\mathfrak{m})/k$ and k'_i/k are separable, k perfect (3.17.106), (3.33.29), so a finite product of fields (3.33.12), as in the statement, applied to k'_i in place of k'), to a prime of that ring, automatically maximal (3.33.13) so $\mathfrak{p}_{i,j}$ is a maximal ideal of $\mathcal{O}_{k'_i}$. By the finite case just established (taking k'_i for k'), $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ is a DVR of $F_{k'_i}$, with $\mathcal{O} \subsetneq \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ and π generating $\mathfrak{p}_{i,j}\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$.

Claim: $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$. For $k'_i \subseteq k'_l$, $\mathfrak{p}_{l,j} \cap \mathcal{O}_{k'_i} = \mathfrak{p}_j \cap \mathcal{O}_{k'_i} = \mathfrak{p}_{i,j}$, so (3.5.9) gives $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}} \subseteq \left(\mathcal{O}_{k'_l}\right)_{\mathfrak{p}_{l,j}}$: a directed system. As $\mathcal{O}_{k'} = \bigcup_i \mathcal{O}_{k'_i}$ is a directed union of subrings with $\mathfrak{p}_{i,j} = \mathfrak{p}_j \cap \mathcal{O}_{k'_i}$, (3.5.10) gives $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$.

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Claim: $k' \subsetneq \mathcal{O}_j \subsetneq F_{k'}$. As π generates the maximal ideal of every $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$, (3.29.6) applies directly (with $F_{k'} = \bigcup_i F_{k'_i}$ the corresponding directed union of fields): $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ is a valuation ring of $F_{k'}$ with maximal ideal $(\pi)\mathcal{O}_j$, and $\mathcal{O}_j \subsetneq F_{k'}$. Also $k' \subsetneq \mathcal{O}_j$: $k' = \bigcup_i k'_i \subseteq \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}} = \mathcal{O}_j$ is immediate, while $\pi \in \mathcal{O}_j \setminus k'$, as $\pi \notin k'_i$ for every i (else $\pi \in \mathfrak{p}_{i,j} \cap k'_i$ would be a unit of $\mathcal{O}_{k'_i}$ (nonzero elements of the subfield $k'_i \subset \mathcal{O}_{k'_i}$ are units), contradicting $\pi \notin \left(\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}\right)$), so $\pi \notin k'$.

This closes b): $\mathcal{O}_j$ is a valuation ring of $F_{k'}$ with $k' \subsetneq \mathcal{O}_j \subsetneq F_{k'}$ – that is, a place of $F_{k'}$ – and π generates both $\mathfrak{m}$ and $\mathfrak{p}_j\mathcal{O}_j$ (above): the same uniformiser generating the maximal ideal on both sides of $\mathcal{O} \subset \mathcal{O}_j$ is exactly what it means for $\mathfrak{m}$ to be unramified in $\mathcal{O}_{k'}$ – matching the factorisation $\mathfrak{b} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ above. When k'/k is finite, $\mathcal{O}_j$ is moreover a discrete valuation ring (d)), recovering the usual ramification index $e(\mathcal{O}_j/\mathcal{O}) = v_{\mathcal{O}_j}(\pi) = 1$.

c) Every valuation ring R with $R \cap F = \mathcal{O}$ is some $\mathcal{O}_j$. Let R be a valuation ring of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$, and $R \cap F = \mathcal{O}$. As $\mathcal{O}_{k'}$ is generated as a ring by (the images of) $\mathcal{O}$ and k' (spanned by simple tensors $a \otimes b = a \cdot b$), both contained in R, $\mathcal{O}_{k'} \subset R$. So $\mathfrak{q} := \mathfrak{m}_R \cap \mathcal{O}_{k'}$ is prime (a contraction of the prime $\mathfrak{m}_R$ along $\mathcal{O}_{k'} \hookrightarrow R$); it contains $\mathfrak{b}$, as $\pi \in \mathfrak{m}_R$ (else $\pi^{-1} \in R$, so $\pi^{-1} \in R \cap F = \mathcal{O}$, contradicting π a non-unit of $\mathcal{O}$), giving $\mathfrak{b} = \pi\mathcal{O}_{k'} \subset \mathfrak{m}_R \cap \mathcal{O}_{k'} = \mathfrak{q}$. So $\mathfrak{q}$ corresponds to a prime, automatically maximal (3.33.13), of the finite-dimensional ring $\mathcal{O}_{k'}/\mathfrak{b} \cong \prod_j E_j$; so $\mathfrak{q} = \mathfrak{p}_j$ for some j.

Every $s \in \mathcal{O}_{k'} \setminus \mathfrak{p}_j$ then satisfies $s \notin \mathfrak{q} = \mathfrak{m}_R \cap \mathcal{O}_{k'}$, that is $s \notin \mathfrak{m}_R$, so s is a unit of R; by the universal property of localisation, $\mathcal{O}_j$ then embeds in R. As $\mathfrak{m}_{\mathcal{O}_j} = \pi\mathcal{O}_j$ and $\pi \in \mathfrak{m}_R$, $\mathfrak{m}_{\mathcal{O}_j} \subseteq \mathfrak{m}_R \cap \mathcal{O}_j$; conversely $\mathfrak{m}_R \cap \mathcal{O}_j$ is a proper ideal of the local ring $\mathcal{O}_j$, hence $\subseteq \mathfrak{m}_{\mathcal{O}_j}$: so $\mathfrak{m}_{\mathcal{O}_j} = \mathfrak{m}_R \cap \mathcal{O}_j$, that is R dominates $\mathcal{O}_j$ (3.23.4). As $\mathcal{O}_j$ is a valuation ring, it is maximal under domination (3.23.13), so $R = \mathcal{O}_j$.

e) $\mathcal{O}_j$ restricts to $\mathcal{O}$. As $\mathcal{O}_{k'} \subset F_{k'}$ is a domain, $\mathcal{O}_j$ is the subring of $F_{k'}$ of fractions a/s with $a \in \mathcal{O}_{k'}$, $s \in \mathcal{O}_{k'} \setminus \mathfrak{p}_j$, so $\mathcal{O} \subset \mathcal{O}_{k'} \subset \mathcal{O}_j$. Moreover $\mathfrak{m} \subseteq \mathfrak{m}\mathcal{O}_{k'} = \mathfrak{b} \subseteq \mathfrak{p}_j$ (a)), so $\mathfrak{m} \subseteq \mathfrak{p}_j\mathcal{O}_j = \mathfrak{m}_{\mathcal{O}_j}$ (3.5.42): and $\mathfrak{p}_j\mathcal{O}_j$ is a proper ideal. Since $\mathcal{O}$ is a valuation ring of F, (3.23.6) (with $B = \mathcal{O}_j$, $\mathfrak{n} = \mathfrak{p}_j\mathcal{O}_j$) gives $\mathcal{O}_j \cap F = \mathcal{O}$ and $\mathfrak{p}_j\mathcal{O}_j \cap F = \mathfrak{m}$. Combined with c) this shows that the valuation rings R of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$ and $R \cap F = \mathcal{O}$ are precisely $\mathcal{O}_1, \ldots, \mathcal{O}_r$. ∎

Remark 3.33.49 (Relation to the Stacks Project)
For k'/k finite*, this result also follows by combining two general facts from [Stacks]: unramified ring maps are stable under base change ([Stacks] (Tag 00US), applied to the finite separable, hence unramified, extension $k \to k(\mathfrak{m})$, base-changed along $k \to k'$), and a finite-dimensional algebra over a field splits, after any base change, into a product of local factors dual to its maximal ideals (the Stacks Project analogue of (3.33.16), (3.33.13) above).*

For k'/k merely algebraic*, the Stacks Project's treatment of unramified morphisms is developed for morphisms of finite type, so does not directly cover this case; the passage to infinite k' via the directed union of finite subextensions, as in* In general*, $\mathcal{O}_j$ is a valuation ring above, has no immediate analogue there and is instead the kind of limit argument found in the Stacks Project's chapter on limits of rings (e.g. [Stacks] (Tag 07RN ff.), on writing a ring as a filtered colimit and transporting properties of the terms to the colimit).*

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